

PROBLEM SET 3

ECON301—ECONOMETRICS I

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Instructions: Due on 20/03/2022; scan or type your work and submit it through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Let $\{W_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with $E(W_i) = 0$ and $\text{Var}(W_i) = \sigma_W^2$, show that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \rightarrow_d N(0, \sigma_W^2)$$

2. Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of random variables such that $E(X_n) \rightarrow c$ and $\text{Var}(X_n) \rightarrow 0$ as $n \rightarrow \infty$. Show that then $X_n \rightarrow_P c$.
3. **Chebychev's second inequality.** Let X be a random variable with $E(X) = \mu < \infty$ and $\text{Var}(X) = \sigma^2$ with $0 < \sigma^2 < \infty$. Then for any real number $k > 0$,

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Prove this result.

4. Suppose $\bar{X}$ is the mean of 100 observations from a population with mean μ and variance $\sigma^2 = 9$. Find limits between which $\bar{X} - \mu$ will lie with probability at least 0.90 using:

- (a) The second Chebychev's inequality from item 3.
- (b) The Central Limit Theorem.

Comment on your results. Which limits are tighter?

5. Let $\{Y_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. random variables with Bernoulli probability mass functions with unknown parameter p . Let $\hat{p}_n = \frac{1}{n} \sum_{i=1}^n Y_i$.

- (a) Show that $\hat{p}_n$ is a consistent estimator of p .
- (b) Find and justify the large sample distribution of

$$\frac{\hat{p}_n - p}{\sqrt{\frac{p(1-p)}{n}}}$$

- (c) Find and justify a consistent estimator of $\text{Var}(Y_1)$.
- (d) Find and justify the large sample distribution of

$$\frac{\hat{p}_n - p}{\sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}}}$$

- (e) A sample of size 127 is such that $\sum_{i=1}^{127} Y_i = 39$. Find a 99% confidence interval for p .
- (f) Suppose we are interested in the *odds ratio*, $\gamma = p/(1-p)$, which is the probability of success over the probability of failure. Propose a consistent estimator of γ and show that it is indeed consistent. Call your proposed estimator $\hat{\gamma}_n$.
- (g) Consider the following theorem:

Theorem (Delta Method). If $\sqrt{n}(T_n - \tau) \rightarrow_d N(0, \psi^2)$ and $U_n = h(T_n)$ where h is continuously differentiable at τ , then

$$\sqrt{n}(U_n - h(\tau)) \rightarrow_d N(0, [h'(\tau)]^2 \psi^2).$$

Equivalently, if $T_n \sim AN(\tau, \psi^2/n)$ and $U_n = h(T_n)$ is continuously differentiable at τ , then

$$U_n \sim AN\left(h(\tau), \frac{[h'(\tau)]^2 \psi^2}{n}\right).$$

Use the Delta Method to show that $\sqrt{n}(\hat{\gamma}_n - \gamma) \rightarrow_d N(0, p/(1-p)^3)$.

- (h) Propose a consistent estimator of $p/(1-p)^3$ and show that it is indeed consistent. Call your proposed estimator s_n^2 .
- (i) Give the details of building an asymptotic pivot to make inference on γ .
- (j) Construct an asymptotic $(1-\alpha)\%$ confidence interval for γ .

6. The lifetime of a certain computer chip that a company manufactures is characterized by the population distribution

$$f(z; \theta) = \frac{1}{\theta} e^{-z/\theta} I_{(0, \infty)}(z),$$

where z is measured in thousand of hours. Let $X_1, \dots, X_n$ represent and i.i.d. random variables with the density $f(z; \theta)$. An outcome of the random variable

$$Y_n = \frac{1}{n} \left(1 + \sum_{i=1}^n X_i \right)$$

will be used to provide an estimate of θ .

- (a) Is it true that $Y_n \rightarrow_{L_2} \theta$? Is it true that $Y_n \rightarrow_P \theta$?
 - (b) Define an asymptotic distribution for Y_n .
 - (c) Suppose that $n = 100$, and $\theta = 10$. use the asymptotic distribution you defined in (b) to approximate the probability that $Y_n \geq 15$.
7. Let $X_1, \dots, X_n$ be i.i.d. random variables having continuous uniform distributions of the form $f(z) = I_{[0,1]}(z)$.
- (a) Define an asymptotic distribution for $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.
 - (b) Using your result from (a), argue that

$$\left(\sum_{i=1}^{12} X_i \right) - 6 \approx Z \sim N(0, 1).$$